THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 2	$-4x + 2y = 6$ becomes $-4x + 2y - 6 = 0$.
2	**(a)** Yes	$5(5) + (7)(3) = 46$, equal to 46.
3	**(b)** 5	$6y = 34 - (2)(2)$ gives $y = 5$.
4	**(d)** a straight line	Fact.
5	**(c)** 4	Compare with $y = mx + c$.
6	**(c)** parallel (or the same line)	Equal slopes.
7	**(a)** 4	$1(2) + (7)k = 30$ gives $k = 4$.
8	**(b)** $\frac{14}{3}$	Put x=0: $y = \frac{14}{3}$.
9	**(b)** 4	$4x - 5y = 1$ becomes $4x - 5y - 1 = 0$.
10	**(d)** No	$6(7) + (5)(4) = 62$, not equal to 40.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$4x + 8y - 5 = 0$ (1); $a = 4$, $b = 8$, $c = -5$ (1).	1+1
12	Left side $= 7(4) + (2)(2) = 32$; Right side $= 32$ (1); Yes, it is a solution (1).	1+1
13	Substitute $x = 4$: $4(4) + (7)k = 44$; $7k = 28$ (1); $k = 4$ (1).	1+1
14	$-9x + 9y - 6 = 0$ (1); $a = -9$, $b = 9$, $c = -6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 0$: $6y = 36$ gives $y = 6$, so $(0, 6)$ (1); $y = 0$: $4x = 36$ gives $x = 9$, so $(9, 0)$ (2).	1+2
16	Slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - (4)}{0 - (1)}$ (1); $= 2$ (2).	1+2
17	Left side $= 3(6) + (2)(2) = 22$ (1); Right side $= 22$ (1); Yes, it is a solution (1).	1+1+1
	OR: $x = 0$: $5y = 15$ gives $y = 3$, so $(0, 3)$ (1); $y = 0$: $5x = 15$ gives $x = 3$, so $(3, 0)$ (2).	1+2
18	Slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - (1)}{4 - (0)}$ (1); $= \frac{3}{4}$ (2).	1+2
	OR: Left side $= 7(7) + (7)(7) = 98$ (1); Right side $= 98$ (1); Yes, it is a solution (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$6y = -7x + 43$ (1); $y = -\frac{7}{6}x + \frac{43}{6}$ (2); Slope $= -\frac{7}{6}$, y-intercept $= \frac{43}{6}$ (2).	1+2+2
	OR: $y = 11x + 31$ (1); For $x = 4$: $y = 11 \times 4 + 31$ (2); Fare ₹75 (2).	1+2+2
20	$6y = -x + 28$ (1); $y = -\frac{1}{6}x + \frac{14}{3}$ (2); Slope $= -\frac{1}{6}$, y-intercept $= \frac{14}{3}$ (2).	1+2+2
	OR: $y = 11x + 45$ (1); For $x = 9$: $y = 11 \times 9 + 45$ (2); Fare ₹144 (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3	$6y = 42 - (6)(4)$ gives $y = 3$.
2	**(c)** infinitely many solutions	Each point on the line.
3	**(c)** 8	Compare with $y = mx + c$.
4	**(a)** a vertical line	All points have $x = 5$.
5	**(c)** 2	$3(1) + (1)k = 5$ gives $k = 2$.
6	**(c)** $\frac{10}{3}$	Put y=0: $x = \frac{10}{3}$.
7	**(a)** 3	$3x - 6y = 6$ becomes $3x - 6y - 6 = 0$.
8	**(a)** Yes	$7(4) + (2)(1) = 30$, equal to 30.
9	**(d)** 5	$4y = 45 - (5)(5)$ gives $y = 5$.
10	**(b)** the origin	$(0, 0)$ satisfies it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Left side $= 5(1) + (7)(6) = 47$; Right side $= 38$ (1); No, it is not a solution (1).	1+1
12	Substitute $x = 3$: $7(3) + (6)k = 33$; $6k = 12$ (1); $k = 2$ (1).	1+1
13	$-2x - 3y - 8 = 0$ (1); $a = -2$, $b = -3$, $c = -8$ (1).	1+1
14	Left side $= 2(4) + (7)(1) = 15$; Right side $= 15$ (1); Yes, it is a solution (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 0$: $4y = 60$ gives $y = 15$, so $(0, 15)$ (1); $y = 0$: $6x = 60$ gives $x = 10$, so $(10, 0)$ (2).	1+2
16	Slope $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{6 - (4)}{6 - (0)}$ (1); $= \frac{1}{3}$ (2).	1+2
17	Left side $= 3(6) + (4)(6) = 42$ (1); Right side $= 42$ (1); Yes, it is a solution (1).	1+1+1
	OR: $x = 0$: $2y = 28$ gives $y = 14$, so $(0, 14)$ (1); $y = 0$: $4x = 28$ gives $x = 7$, so $(7, 0)$ (2).	1+2
18	Slope $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{0 - (2)}{5 - (2)}$ (1); $= -\frac{2}{3}$ (2).	1+2
	OR: Left side $= 1(4) + (3)(6) = 22$ (1); Right side $= 26$ (1); No, it is not a solution (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$5y = -7x + 33$ (1); $y = -\frac{7}{5}x + \frac{33}{5}$ (2); Slope $= -\frac{7}{5}$, y-intercept $= \frac{33}{5}$ (2).	1+2+2
	OR: $y = 10x + 55$ (1); For $x = 6$: $y = 10 \times 6 + 55$ (2); Fare ₹115 (2).	1+2+2
20	$7y = -3x + 19$ (1); $y = -\frac{3}{7}x + \frac{19}{7}$ (2); Slope $= -\frac{3}{7}$, y-intercept $= \frac{19}{7}$ (2).	1+2+2
	OR: $y = 19x + 27$ (1); For $x = 8$: $y = 19 \times 8 + 27$ (2); Fare ₹179 (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Two Variables, One Line** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2	Compare with $y = mx + c$.
2	**(c)** a horizontal line	All points have $y = -3$.
3	**(b)** 3	$1(1) + (2)k = 7$ gives $k = 3$.
4	**(b)** 7	Put x=0: $y = 7$.
5	**(d)** 2	$2x + 2y = 2$ becomes $2x + 2y - 2 = 0$.
6	**(d)** Yes	$2(5) + (-1)(3) = 7$, equal to 7.
7	**(d)** 4	$6y = 44 - (5)(4)$ gives $y = 4$.
8	**(a)** its steepness	Rise over run.
9	**(b)** -1	Compare with $y = mx + c$.
10	**(d)** its coordinates satisfy the equation of the line	Definition.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substitute $x = 4$: $5(4) + (4)k = 40$; $4k = 20$ (1); $k = 5$ (1).	1+1
12	$-9x + 4y - 5 = 0$ (1); $a = -9$, $b = 4$, $c = -5$ (1).	1+1
13	Left side $= 7(5) + (6)(4) = 59$; Right side $= 59$ (1); Yes, it is a solution (1).	1+1
14	Substitute $x = 5$: $3(5) + (2)k = 19$; $2k = 4$ (1); $k = 2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$x = 0$: $5y = 35$ gives $y = 7$, so $(0, 7)$ (1); $y = 0$: $5x = 35$ gives $x = 7$, so $(7, 0)$ (2).	1+2
16	Slope $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{6 - (0)}{4 - (3)}$ (1); $= 6$ (2).	1+2
17	Left side $= 2(5) + (7)(5) = 45$ (1); Right side $= 33$ (1); No, it is not a solution (1).	1+1+1
	OR: $x = 0$: $4y = 44$ gives $y = 11$, so $(0, 11)$ (1); $y = 0$: $4x = 44$ gives $x = 11$, so $(11, 0)$ (2).	1+2
18	Slope $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{4 - (2)}{2 - (4)}$ (1); $= -1$ (2).	1+2
	OR: Left side $= 2(7) + (5)(5) = 39$ (1); Right side $= 27$ (1); No, it is not a solution (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$7y = -5x + 48$ (1); $y = -\frac{5}{7}x + \frac{48}{7}$ (2); Slope $= -\frac{5}{7}$, y-intercept $= \frac{48}{7}$ (2).	1+2+2
	OR: $y = 16x + 56$ (1); For $x = 5$: $y = 16 \times 5 + 56$ (2); Fare ₹136 (2).	1+2+2
20	$6y = -2x + 28$ (1); $y = -\frac{1}{3}x + \frac{14}{3}$ (2); Slope $= -\frac{1}{3}$, y-intercept $= \frac{14}{3}$ (2).	1+2+2
	OR: $y = 13x + 49$ (1); For $x = 9$: $y = 13 \times 9 + 49$ (2); Fare ₹166 (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.